THE UNIVERSAL ASKEY-WILSON ALGEBRA MAPS ONTO THE ASKEY-WILSON ALGEBRA, SENDING ALPHA, BETA, GAMMA TO A FIXED CENTRAL SCALAR

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Allowed standard prerequisites (editorial).

- The evaluation identities for the generators A, B, C of the Askey-Wilson algebra Lambda. Same source, Lemmas labelled lem:xyz and lem:multtable2; invoked inside the counted proof at lines 948-964. Supply the three displayed identities satisfied by the images A^vee, B^vee, C^vee.
- The defining relations of the universal Askey-Wilson algebra. Same source, Definition labelled def:uaw; invoked inside the counted proof at lines 965-971. Turns the three identities into the relations that guarantee existence of the homomorphism.
- The definitions of alpha, beta, gamma and of the scalar mu. Same source, Definition labelled def:abc; invoked inside the counted proof at lines 974-978. Identifies the images of alpha, beta, gamma with nu I, where nu = (q^2+q^{-2}) mu + mu^2.

Source reference. Original definition 1.3: sources/sigma-2011-069-askey-wilson/pinned-original-source.tex, source lines 977–977. Navigation pointer for \ref{def:abc} used inside the counted span (line 977); the argument that consumes it is retained in the span.

Date: doi:10.3842/SIGMA.2011.069; CC BY4.0.

Source reference. Original definition 1.2: sources/sigma-2011-069-askey-wilson/pinned-original-source.tex, source lines 971–971. Navigation pointer for \ref{def:uaw} used inside the counted span (line 971); the argument that consumes it is retained in the span.

Source reference. Original lemma 1.12: sources/sigma-2011-069-askey-wilson/pinned-original-source.tex, source lines 949–949. Navigation pointer for \ref{lem:multtable2} used inside the counted span (line 949); the argument that consumes it is retained in the span.

Source reference. Original lemma 1.11: sources/sigma-2011-069-askey-wilson/pinned-original-source.tex, source lines 948–948. Navigation pointer for \ref{lem:xyz} used inside the counted span (line 948); the argument that consumes it is retained in the span.

Lemma 0.1. *There exists a unique* $\mathbb{F}$*-algebra homomorphism* $\pi : \Delta \to \Lambda$ *that sends*

$$A \mapsto q\mathcal{A} + q^{-1}\mathcal{A}^{-1}, \qquad B \mapsto q\mathcal{B} + q^{-1}\mathcal{B}^{-1}, \qquad C \mapsto q\mathcal{C} + q^{-1}\mathcal{C}^{-1}.$$

The homomorphism π *sends*

$$\alpha \mapsto \nu I, \qquad \beta \mapsto \nu I, \qquad \gamma \mapsto \nu I, \tag{1}$$

where $\nu = (q^2 + q^{-2})\mu + \mu^2$.

Proof. Define

$$A^\vee = q\mathcal{A} + q^{-1}\mathcal{A}^{-1}, \qquad B^\vee = q\mathcal{B} + q^{-1}\mathcal{B}^{-1}, \qquad C^\vee = q\mathcal{C} + q^{-1}\mathcal{C}^{-1}.$$

By Lemma 1.11 and Lemma 1.12,

$$(q + q^{-1})A^\vee + \frac{qB^\vee C^\vee - q^{-1}C^\vee B^\vee}{q - q^{-1}} = \nu I, \tag{2}$$

$$(q + q^{-1})B^\vee + \frac{qC^\vee A^\vee - q^{-1}A^\vee C^\vee}{q - q^{-1}} = \nu I, \tag{3}$$

$$(q + q^{-1})C^\vee + \frac{qA^\vee B^\vee - q^{-1}B^\vee A^\vee}{q - q^{-1}} = \nu I. \tag{4}$$

By (2)–(4) and since νI is central, the elements $A^\vee$, $B^\vee$, $C^\vee$ satisfy the defining relations for Δ from Definition 1.2. Therefore the homomorphism π exists. The homomorphism π is unique since A, B, C generate Δ. Line (1) follows from Definition 1.3 and (2)–(4). □

References

[1] Alperin R.C., Notes: $\mathrm{PSL}_2(\mathbb{Z}) = \mathbb{Z}_2 \star \mathbb{Z}_3$, *Amer. Math. Monthly* **100** (1993), 385–386.